

Lecture 4 - Consumer Choice

ECON 3070 - Intermediate Microeconomic Theory

Danielle Parks

In the previous lecture, we looked at consumer preferences.

- We considered how consumers ranked various bundles of goods.
- We saw how to graphically and numerically represent these preferences.

In the next lecture, we will consider how consumers choose which bundle to consume.

The Budget Constraint

Consumers only have a limited amount of money to spend on goods and services.

- A consumer's **budget constraint** defines the set of baskets that a consumer can purchase.
- If P_x is the price of good x , then $P_x x$ is the amount of money spent on good x
- Similarly, $P_y y$ is the amount of money spent on good y .

Because a consumer can only spend as much as their income, I , their budget constraint can be expressed as:

$$P_x x + P_y y \leq I$$

Budget Constraint

The budget constraint can be generalized to more than two goods:

$$P_{x_1}x_1 + P_{x_2}x_2 + \dots + P_{x_n}x_n \leq I$$

- Because more is better (remember our assumptions), the consumer will always use up their whole budget (I).
- In general, however, a consumer can purchase any bundle which costs I \$ or less.

The Budget Line

A consumer's budget constraint for two goods can also be represented graphically. In the equation $P_x x + P_y y = I$, solve for y .

The rearranged equation is

$$y = \frac{I}{P_y} - \frac{P_x}{P_y}x$$

- This equation in slope-intercept form can be used to plot our budget line.

A **budget line** shows all combinations of goods x and y that can be purchased for a given budget I , at given prices P_x and P_y .

The Budget Line

Try It Yourself

Write the budget constraint with $I = 800$, $P_X = 20$, and $P_Y = 40$. Then, solve for y .

Solution

The budget constraint is

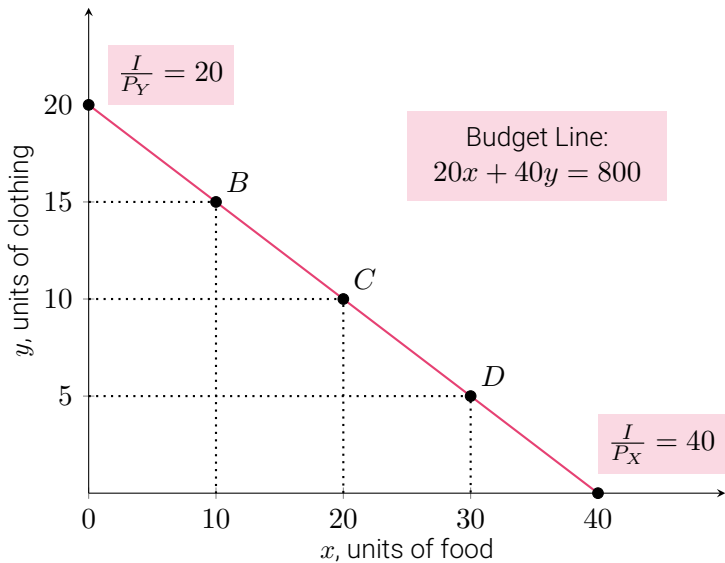
$$20x + 40y = 800$$

Solving for y :

$$40y = 800 - 20x \implies y = 20 - \frac{1}{2}x$$

This is exactly the budget line plotted on the next slide – notice the y -intercept of 20 and slope of $-1/2$.

Figure: The Budget Line



The Budget Line

The y intercept tells us how many units of good y can be purchased if none of good x is bought. Similarly for the x intercept.

The slope tells us how many units of good y must be given up to obtain one more unit of good x .

Conceptual questions

- Remember that a change in income shifts the budget line, without changing the slope (why?)
- What would a change in the price of one (or both) of the goods do?

The Budget Line

Try It Yourself

Suppose that Sammy has \$200 in disposable income to spend each month on concert tickets and football tickets. Concert tickets cost \$50 and football tickets cost \$20. The price of concert tickets increases by \$25, and the price of football tickets increases by \$10. Assume that Sammy's disposable income doesn't change.

Draw the budget line before and after.

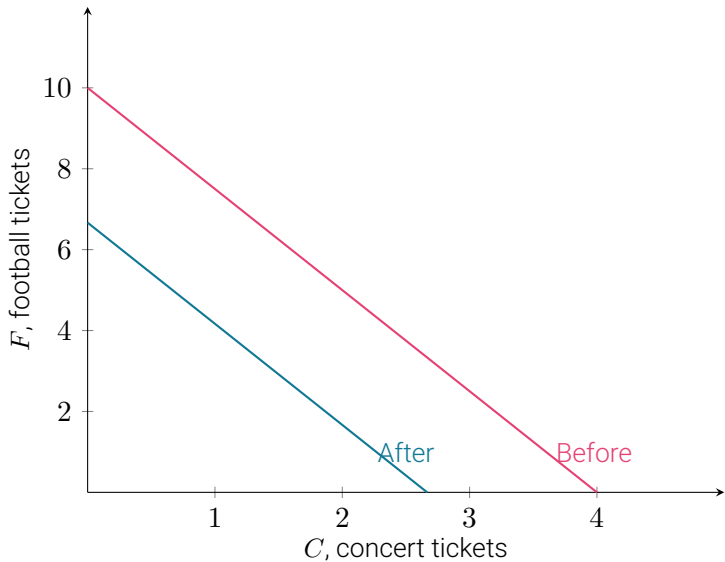
Solution

Before: $50C + 20F = 200$. Intercepts: $C = 4$ (if $F = 0$), $F = 10$ (if $C = 0$). Slope $= -50/20 = -2.5$.

After: concert tickets now cost \$75, football tickets now cost \$30. So $75C + 30F = 200$. Intercepts: $C = 200/75 \approx 2.67$, $F = 200/30 \approx 6.67$. Slope $= -75/30 = -2.5$.

Key insight: both prices rose by exactly 50% ($50 \rightarrow 75$, $20 \rightarrow 30$), so the price *ratio* – and therefore the slope – didn't change. The budget line shifts **inward, parallel** to the original line, just like a decrease in real income.

Figure: Sammy's Budget Line, Before and After



Optimal Choice

Now that we know the consumer's problem (maximize utility), and their constraint (the budget constraint), we can determine the consumer's optimal choice.

- The **optimal choice** is the amount of each good that should be purchased in order to maximize utility.
- This basket of goods must be located on the budget line (why not below?).

More formally, we can state the consumer's **utility maximization** problem as:

$$\max_{(x,y)} U(x, y) \text{ subject to: } P_x x + P_y y \leq I$$

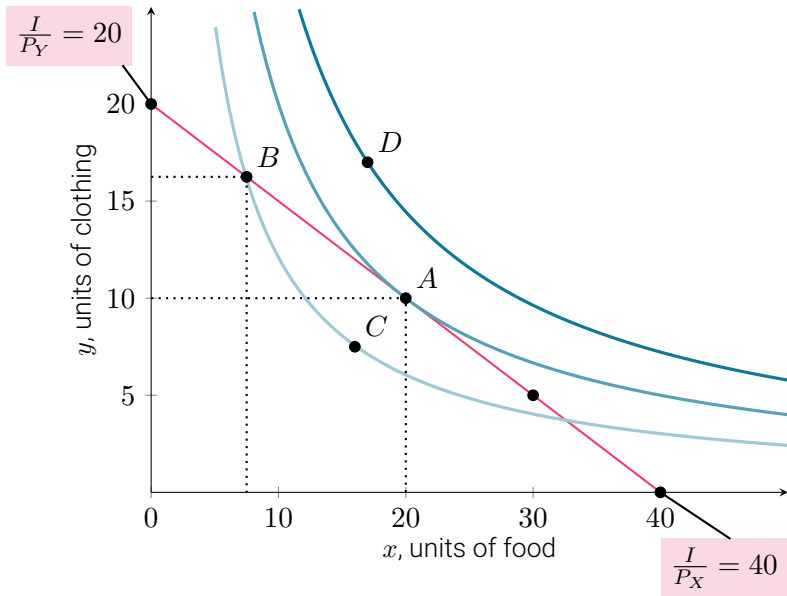
Utility Maximization

How can we interpret this graphically?

- Remember that the consumer wants to get to the highest indifference curve possible.
- That means moving as far up and to the right as possible
- But the consumer can't move beyond their budget line.

The optimal bundle ends up being (in most cases) the point where the budget line is tangent with one of the indifference curves.

Figure: Utility Maximization



Utility Maximization

To find the consumer's optimal level of consumption of x and y , we need some calculus.

We will maximize utility in two-ways:

1. The method of Lagrange
2. 'plugging-in' the constraint to our problem.

Method of Lagrange

Define the Lagrangian ($\mathcal{L}$) as

$$\mathcal{L}(x, y, \lambda) = U(x, y) + \lambda(I - P_x x - P_y y)$$

Maximizing the Lagrangian with respect to x , y , and λ is equivalent to maximizing the original problem.

Method of Lagrange

To maximize the Lagrangian, take derivatives with respect to x , y , and λ , and set them equal to zero. The first order conditions for an optimum are:

$$\frac{\partial \mathcal{L}}{\partial x} = 0 \Rightarrow \frac{\partial U(x, y)}{\partial x} = \lambda P_x$$

$$\frac{\partial \mathcal{L}}{\partial y} = 0 \Rightarrow \frac{\partial U(x, y)}{\partial y} = \lambda P_y$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 0 \Rightarrow I - P_x x - P_y y = 0$$

Method of Lagrange

Remember that $\frac{\delta U(x,y)}{\delta x} = MU_x$ and $\frac{\delta U(x,y)}{\delta y} = MU_y$. Then, rearranging terms:

$$\frac{MU_x}{P_x} = \lambda \quad \text{and} \quad \frac{MU_y}{P_y} = \lambda$$

We can combine the first two equations and rearrange terms, and the result is:

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y}$$

The resulting equality is our [optimality condition](#).

Example

Consider $U(x, y) = 2xy$ and $I = 10, P_x = 2, P_y = 1$. Solve for the Consumers optimal choices of goods x and y using the Lagrangian Method.

Method of Lagrange

Try It Yourself

Consider $U(x, y) = xy$ and $I = 80, P_x = 4, P_y = 2$. Solve for the Consumers optimal choices of goods x and y using the Lagrangian Method.

Solution

The Lagrangian is

$$\mathcal{L}(x, y, \lambda) = xy + \lambda(80 - 4x - 2y)$$

First order conditions:

$$\frac{\delta \mathcal{L}}{\delta x} = y - 4\lambda = 0 \implies y = 4\lambda$$

$$\frac{\delta \mathcal{L}}{\delta y} = x - 2\lambda = 0 \implies x = 2\lambda$$

$$\frac{\delta \mathcal{L}}{\delta \lambda} = 80 - 4x - 2y = 0$$

Solution (continued)

From the first two conditions, $y = 4\lambda$ and $x = 2\lambda$, so $y = 2x$.

Plugging into the budget constraint:

$$4x + 2(2x) = 80 \implies 8x = 80 \implies x^* = 10$$

Then $y^* = 2x^* = 20$.

Optimal bundle: $x^* = 10, y^* = 20$.

Unconstrained Optimization

To solve using unconstrained optimization, first solve for y in the budget constraint $P_x x + P_y y = I$.

$$y = \frac{I - P_x x}{P_y}$$

We then plug this into our utility function, and our problem becomes:

$$\max_x U\left(x, \frac{I - P_x x}{P_y}\right)$$

Unconstrained Optimization

How do we maximize (or minimize) any function?

- Take derivatives, and set them equal to zero.

Taking the total derivative of $U(x, \frac{I - P_x x}{P_y})$ w.r.t x (using the chain-rule!):

$$\begin{aligned}\frac{dU}{dx} &= \frac{\delta U}{\delta x} + \frac{\delta U}{\delta y} \frac{dy}{dx} = 0 \\ \Rightarrow \frac{\delta U}{\delta x} - \frac{\delta U}{\delta y} \frac{P_x}{P_y} &= 0 \\ \Rightarrow \frac{\delta U}{\delta x} / P_x &= \frac{\delta U}{\delta y} / P_y \\ \rightarrow \frac{MU_x}{P_x} &= \frac{MU_y}{P_y}\end{aligned}$$

Unconstrained Optimization

Example

Consider $U(x, y) = 2xy$ and $I = 10, P_x = 2, P_y = 1$. Solve for the Consumers optimal choices of goods x and y using Unconstrained Optimization.

Solution

Solve the budget constraint for y :

$$y = \frac{I - P_x x}{P_y} = \frac{10 - 2x}{1} = 10 - 2x$$

Plug into the utility function:

$$U(x) = 2x(10 - 2x) = 20x - 4x^2$$

Take the derivative and set it to zero:

$$\frac{dU}{dx} = 20 - 8x = 0 \implies x^* = 2.5$$

Then $y^* = 10 - 2(2.5) = 5$.

Unconstrained Optimization

Try It Yourself

Consider $U(x, y) = xy$ and $I = 80, P_x = 4, P_y = 2$. Solve for the Consumers optimal choices of goods x and y using Unconstrained Optimization.

Solution

Solve the budget constraint for y :

$$y = \frac{80 - 4x}{2} = 40 - 2x$$

Plug into the utility function:

$$U(x) = x(40 - 2x) = 40x - 2x^2$$

Take the derivative and set it to zero:

$$\frac{dU}{dx} = 40 - 4x = 0 \implies x^* = 10$$

Then $y^* = 40 - 2(10) = 20$.

The Optimality Condition

We have just found our first **optimality condition**! Now that we have derived this from first principles, we don't have to do that again!

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y}$$

- The **optimality condition** tells us that the consumer maximizes utility when the marginal utility per dollar spent on both goods is equal.
- Think about what would happen if it weren't equal for the two goods.

The Optimality Condition

We can also rearrange the **optimality condition** as follows:

$$\frac{MU_x}{MU_y} = \frac{P_x}{P_y}$$

- This tells us that the slope of our indifference curve should equal the slope of our budget line.
- The result is that the optimal basket is located where the budget line is exactly tangent to one of the indifference curves.
- Compare this to our graphical result.

The Optimality Condition

So far, we only considered the case where baskets contain two goods. However, the **optimality condition** holds for any number of goods.

$$\frac{MU_{x_1}}{P_{x_1}} = \frac{MU_{x_2}}{P_{x_2}} = \dots = \frac{MU_{x_n}}{P_{x_n}}$$

- As before, the utility per dollar spent on each good (or “bang for your buck”), must be equal.

This is a *key insight*! This **optimality condition** is how we think individuals balance **trade-offs** on what to buy

The Optimality Condition

Try It Yourself

Consider a market with only two goods, bread and cheese. Suppose that at Francis' current level of consumption, his marginal utility of bread is 4, and his marginal utility of cheese is 6. Suppose that a loaf of bread costs \$3, and a pound of cheese costs \$6. What should Francis do?

Solution

Compute utility per dollar spent on each good:

$$\frac{MU_{bread}}{P_{bread}} = \frac{4}{3} \approx 1.33 \qquad \frac{MU_{cheese}}{P_{cheese}} = \frac{6}{6} = 1$$

Since $\frac{MU_{bread}}{P_{bread}} > \frac{MU_{cheese}}{P_{cheese}}$, bread gives Francis more utility per dollar than cheese.

Francis should buy more bread and less cheese. As he does, MU_{bread} will fall (diminishing marginal utility) and MU_{cheese} will rise, until the two ratios are equal – at which point he's reached his optimal bundle.

Finding the Optimal Bundle

Now that we know something about the optimal bundle (from the [optimality condition](#)), how do we actually find it?

- Well we also know that the consumer has only $\$I$ to spend.
- This gives us a system of two equations (the [optimality condition](#) and the budget constraint), and two unknowns (quantity consumed of goods x and y).

All that's left is some (sometimes messy) algebra.

Finding the Optimal Bundle

Suppose $U(x, y) = x^{\frac{1}{2}}y^{\frac{1}{2}}$. Let $P_x = 2$, $P_y = 3$ and $I = 24$.

Calculate MU_x and MU_y and the optimality condition

Finding the Optimal Bundle

Suppose $U(x, y) = x^{\frac{1}{2}}y^{\frac{1}{2}}$. Let $P_x = 2$, $P_y = 3$ and $I = 24$.

$$MU_x = \frac{y^{\frac{1}{2}}}{2x^{\frac{1}{2}}} \text{ and } MU_y = \frac{x^{\frac{1}{2}}}{2y^{\frac{1}{2}}}$$

Then our optimality condition is $\frac{y^{\frac{1}{2}}}{2x^{\frac{1}{2}}} / \frac{x^{\frac{1}{2}}}{2y^{\frac{1}{2}}} = \frac{2}{3}$

$$\implies y = \frac{2}{3}x$$

Plugging the optimality condition into our budget constraint

$$2x + 3y = 24 \implies 2x + 3\left(\frac{2}{3}x\right) = 24 \implies x^* = 6$$

Plugging x^* back into the optimality condition, $y^* = \frac{2}{3}6 = 4$

Finding the Optimal Bundle

We could also leave out prices and income, and simply solve for x and y in terms of P_x , P_y , and I .

Take the rearranged **optimality condition** $y = x \frac{P_x}{P_y}$. Plug into budget constraint

$$P_x x + P_y x \left(\frac{P_x}{P_y} \right) = I \implies x^* = \frac{I}{2P_x}$$

Plug x back into the **optimality condition**:

$$y^* = \frac{I}{2P_x} \frac{P_x}{P_y} = \frac{I}{2P_y}$$

Demand Functions

The resulting **demand functions** are:

$$x^*(P_x, P_y, I) = \frac{I}{2P_x} \quad \text{and} \quad y^*(P_x, P_y, I) = \frac{I}{2P_y}$$

The **demand function** is useful since we do the work once and then can predict what that person will consumer under any (P_x, P_y, I) combination for comparative statics

- I'll likely do this on the test

Demand Functions

Try It Yourself

Consider $U(x, y) = x^{\frac{1}{2}}y^{\frac{1}{2}}$ and $I = 80, P_x = 4, P_y = 2$. Solve for the Consumers optimal choices of goods x and y and the resulting demand functions.

Solution

Since $U(x, y) = x^{\frac{1}{2}}y^{\frac{1}{2}}$ has equal exponents, we can use the demand functions derived earlier: $x^* = \frac{I}{2P_x}$ and $y^* = \frac{I}{2P_y}$.

Plugging in $I = 80$, $P_x = 4$, $P_y = 2$:

$$x^* = \frac{80}{2(4)} = \frac{80}{8} = 10 \qquad y^* = \frac{80}{2(2)} = \frac{80}{4} = 20$$

Optimal bundle: $x^* = 10$, $y^* = 20$.

Demand functions: $x^*(P_x, P_y, I) = \frac{I}{2P_x}$, $y^*(P_x, P_y, I) = \frac{I}{2P_y}$.

Three Utility Functions

How do you find demand for the three utility functions?

1. **Cobb-Douglas:** $U(x, y) = x^\alpha y^\beta$
2. **Perfect Substitutes:** $U(x, y) = ax + by$
3. **Perfect Complements:** $U(x, y) = \min(ax, by)$

Cobb-Douglas

$U(x, y) = x^\alpha y^\beta$. Our optimality condition is

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y} \implies \frac{\alpha x^{\alpha-1} y^\beta}{P_x} = \frac{\beta x^\alpha y^{\beta-1}}{P_y} \implies x = \frac{\alpha P_y}{\beta P_x} y$$

Our budget constraint is

$$I = P_x x + P_y y$$

Solve these two equations for x^* and y^* .

Try It Yourself

Consider $U(x, y) = x^{\frac{1}{4}}y^{\frac{3}{4}}$ and $I = 400$, $P_x = 20$, $P_y = 40$. Solve for the Consumers optimal choices of goods x and y and the resulting demand functions.

Solution

Here $\alpha = \frac{1}{4}$ and $\beta = \frac{3}{4}$ (note $\alpha + \beta = 1$). Using the general Cobb-Douglas result:

$$x^*(P_x, P_y, I) = \frac{\alpha I}{P_x} \qquad y^*(P_x, P_y, I) = \frac{\beta I}{P_y}$$

Plugging in $I = 400$, $P_x = 20$, $P_y = 40$:

$$x^* = \frac{\frac{1}{4}(400)}{20} = \frac{100}{20} = 5 \qquad y^* = \frac{\frac{3}{4}(400)}{40} = \frac{300}{40} = 7.5$$

Optimal bundle: $x^* = 5$, $y^* = 7.5$. **Demand functions:** $x^* = \frac{I}{4P_x}$,

$$y^* = \frac{3I}{4P_y}.$$

Perfect Substitutes

$U(x, y) = ax + by$. Our optimality condition is

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y} \implies \frac{a}{P_x} = \frac{b}{P_y}$$

Wait! This isn't always true!!

- e.g. what if $U(x, y) = x + y$, $P_x = 1$, and $P_y = 2$. This says $1 = 1/2$ which isn't true

Since both goods are perfectly substitutable, you should consume **ONLY** the good that provides marginal utility per dollar (bang for your buck)

Perfect Substitutes

$U(x, y) = ax + by$. For perfect substitutes our demand rule is simple:

1. If $MU_x/P_x > MU_y/P_y$:

$$x^* = I/P_x \quad \text{and} \quad y^* = 0$$

2. If $MU_x/P_x < MU_y/P_y$:

$$x^* = 0 \quad \text{and} \quad y^* = I/P_y$$

3. If $MU_x/P_x = MU_y/P_y$:

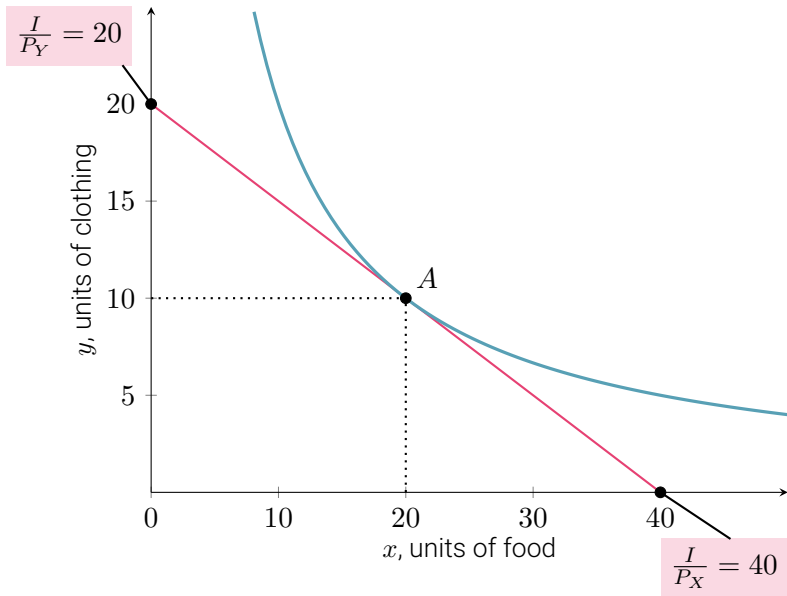
Pick any bundle on your budget line

So far are our maximization problems have had interior solutions where positive quantities of both goods are consumed, e.g. $x^* > 0$ & $y^* > 0$

Key characteristic is that the optimal solution is where the indifference curve is tangent to the budget constraint, which requires preferences to be “well-behaved”:

- Smooth
- Convex (curved in)
- Not touch the two axis

Figure: Utility Maximization

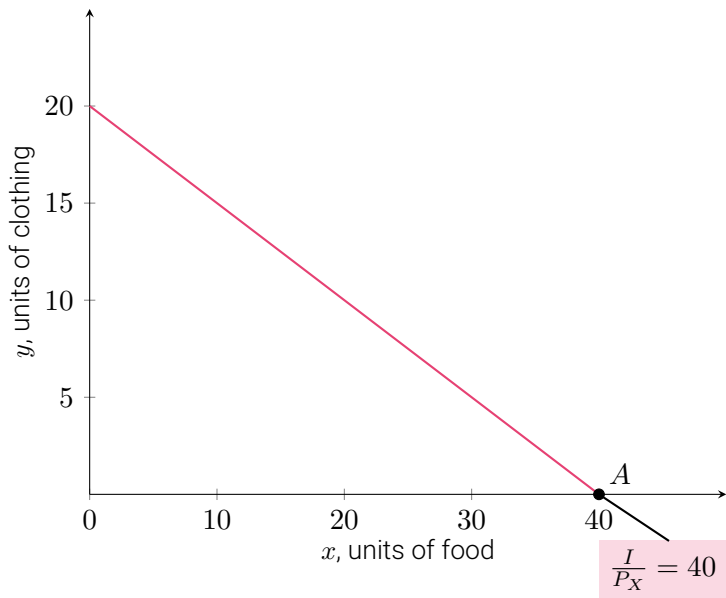


At a corner solution only one good is consumed at a positive quantity,
e.g. $x^* > 0$ & $y^* = 0$

Occurs when preferences are not well-behaved, such as by crossing one of the axis

Optimal solution is to spend all income on one good and none on the other good

Figure: Utility Maximization



Try It Yourself

Consider $U(x, y) = xy + 7y$ and $I = 10, P_x = 1, P_y = 2$. Solve for the Consumers optimal choices of goods x and y and the resulting demand functions.

Solution

This isn't Cobb-Douglas, so we derive marginal utilities directly:

$$MU_x = y \qquad MU_y = x + 7$$

The optimality condition is

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y} \implies \frac{y}{1} = \frac{x+7}{2} \implies x = 2y - 7$$

Solution (continued)

Plug into the budget constraint $x + 2y = 10$:

$$(2y - 7) + 2y = 10 \implies 4y = 17 \implies y^* = \frac{17}{4} = 4.25$$

$$x^* = 2(4.25) - 7 = 1.5$$

Optimal bundle: $x^* = 1.5$, $y^* = 4.25$ (an interior solution, since both are positive).

Demand functions (solving the same way in terms of P_x , P_y , I):

$$x^*(P_x, P_y, I) = \frac{I - 7P_x}{2P_x}$$

$$y^*(P_x, P_y, I) = \frac{I + 7P_x}{2P_y}$$

Perfect Complements

$U(x, y) = \min(ax, by)$. How do we even take a derivative of this?

Instead, recognize that perfect complements means you have to consume always in the proportion $ax = by$.

- If $ax > by$, you are wasting money on x and vice-versa if $by > ax$

$ax = by$ is therefore our **optimality condition**! Our budget constraint is

$$I = P_x x + P_y y$$

Solve these two equations for x^* and y^* .

Perfect Complements

Try It Yourself

Maria always uses 2 scoops of coffee (x) for every 1 cup of milk (y), so her preferences are given by $U(x, y) = \min(2x, y)$.

She has $I = 100$ to spend. Coffee costs $P_x = 4$ per scoop, and milk costs $P_y = 3$ per cup. Solve for her optimal choices of x and y .

Solution

Here $a = 2$ and $b = 1$, so the optimality condition $ax = by$ becomes

$$2x = y$$

Plug into the budget constraint $P_x x + P_y y = I$:

$$4x + 3(2x) = 100 \implies 4x + 6x = 100 \implies 10x = 100 \implies x^* = 10$$

Then $y^* = 2x^* = 20$.

Optimal bundle: $x^* = 10$ scoops of coffee, $y^* = 20$ cups of milk – exactly a 1:2 ratio, as her preferences require.